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Report: Cyclistic Bike-Share Analysis Case Study

Introduction

In this case study, I will perform many real-world tasks of a junior data analyst at a fictional company, Cyclistic. In order to answer the key business questions, I will follow the steps of the data analysis process: Ask, Prepare, Process, Analyze, Share, and Act.

Background

1. Project Introduction:

As a Junior Data Analyst on the Cyclistic marketing analytics team, I am conducting this case study to help drive the company's future growth. To answer our core business questions, I am following the structured data analysis process: **Ask, Prepare, Process, Analyze, Share, and Act**. This report serves as a comprehensive look at my findings and data-driven recommendations.

2. Business Scenario:

The Director of Marketing, Lily Moreno, has identified that Cyclistic's future success depends on maximizing the number of annual memberships. My primary objective is to understand how **casual riders** and **annual members** use Cyclistic bikes differently. By uncovering these insights, I aim to provide the marketing team with the necessary intelligence to design a strategy that converts casual riders into long-term annual members. My recommendations must be approved by the executive team and are therefore backed by compelling data insights and professional visualizations.

3. Organizational Context:

Since its launch in 2016, Cyclistic has grown into a premier bike-share program in Chicago with a fleet of over 5,800 geotracked bicycles and 600 docking stations.

Service Offerings

- I recognize that Cyclistic distinguishes itself through inclusivity, offering not only traditional bikes but also reclining bikes, hand tricycles, and cargo bikes.

- While the majority of our users opt for traditional bikes, approximately 8% utilize assistive options.
- Our data shows that while most riders use the service for leisure, about 30% of our users rely on these bikes for their daily commute.

Pricing and Profitability

Currently, Cyclistic offers three tiers of pricing: **single-ride passes**, **full-day passes**, and **annual memberships**.

- I define "casual riders" as those purchasing single or full-day passes, while "members" are those with annual subscriptions.
- Internal financial analysis has confirmed that annual members are significantly more profitable than casual riders.

The Strategy Shift

Previously, our marketing relied on broad awareness campaigns. However, I am now focusing on Moreno's goal of conversion. Because casual riders are already familiar with our service, they represent a high-potential segment for membership growth. To facilitate this, I have analyzed historical bike trip data to identify the behavioral trends that differentiate these two groups.

Data Analysis Processes

1. Ask

Business Task

Devise marketing strategies to convert casual riders to members.

Analysis Questions

Three questions will guide the future marketing program:

1. How do annual members and casual riders use Cyclistic bikes differently?
2. Why would casual riders buy Cyclistic annual memberships?
3. How can Cyclistic use digital media to influence casual riders to become members?

Moreno has assigned me the first question to answer: How do annual members and casual riders use Cyclistic bikes differently?

2. Prepare

Data Source

I will use Cyclistic's historical trip data to analyze and identify trends from Jan 2024 to Dec 2024 which can be downloaded from [divvy_tripdata](#). The data has been made available by Motivate International Inc. under this [license](#).

This is public data that can be used to explore how different customer types are using Cyclistic bikes. But note that data-privacy issues prohibit using riders' personal information. This means that we won't be able to connect pass purchases to credit card numbers to determine if casual riders live in the Cyclistic service area or if they have purchased multiple single passes.

Data Organization

There are 12 files with naming convention of YYYYMM-divvy-tripdata and each file includes information for one month, such as the ride id, bike type, start time, end time, start station, end station, start location, end location, and whether the rider is a member or not.

The corresponding column names are shown below:

No.	Variable	Description
1	ride_id	Unique ID assigned to each ride
2	rideable_type	classic, docked, or electric
3	started_at	Date and time at the start of trip
4	ended_at	Date and time at the end of trip
5	start_station_name	Name of the station where the ride journey started from
6	start_station_id	ID of the station where the ride journey started from
7	end_station_name	Name of the station where the ride trip ended at
8	end_station_id	ID of the station where the ride trip ended at
9	start_lat	Latitude of starting station
10	start_lng	Longitude of starting station
11	end_lat	Latitude of ending station
12	end_lng	Longitude of ending station
13	member_casual	Type of membership of each rider

and the data type of each variable is depicted below:

Field name	Type	Mode
ride_id	STRING	NULLABLE
rideable_type	STRING	NULLABLE
started_at	TIMESTAMP	NULLABLE
ended_at	TIMESTAMP	NULLABLE
start_station_name	STRING	NULLABLE
start_station_id	STRING	NULLABLE
end_station_name	STRING	NULLABLE
end_station_id	STRING	NULLABLE
start_lat	FLOAT	NULLABLE
start_lng	FLOAT	NULLABLE
end_lat	FLOAT	NULLABLE
end_lng	FLOAT	NULLABLE
member_casual	STRING	NULLABLE

3. Process: Data Cleaning & Transformation

To ensure the data was accurate, consistent, and ready for analysis, I performed the following cleaning and transformation steps using **SQL Server Management Studio 22**:

- **Handling Missing Values:** All rows containing null or missing values across the 12 monthly datasets were identified and removed to maintain data integrity.
- **Data Transformation (New Columns):** I engineered three new columns to enable deeper behavioral analysis:
 - a. **avg_ride_length_in_seconds:** Calculated the difference between `ended_at` and `started_at` to determine trip duration.
 - b. **day_of_week:** Extracted the day from the timestamp to analyze weekly ridership patterns.
 - c. **month:** Extracted the month to identify seasonal trends throughout the 2024 fiscal year.
- **Filtering Outliers:** To prevent skewed results, I excluded rides with a duration of less than one minute (potentially false starts or docking errors) and rides longer than 24 hours (potentially lost or unreturned bikes).
- **Final Dataset Verification:** After these cleaning steps, **4,224,062 rows** were returned for analysis. This process removed a total of **1,496,735 rows** of

inconsistent or incomplete data, ensuring a high-quality foundation for the subsequent Analyze phase.

4. Analyze

Data Analysis:

The analysis question is:

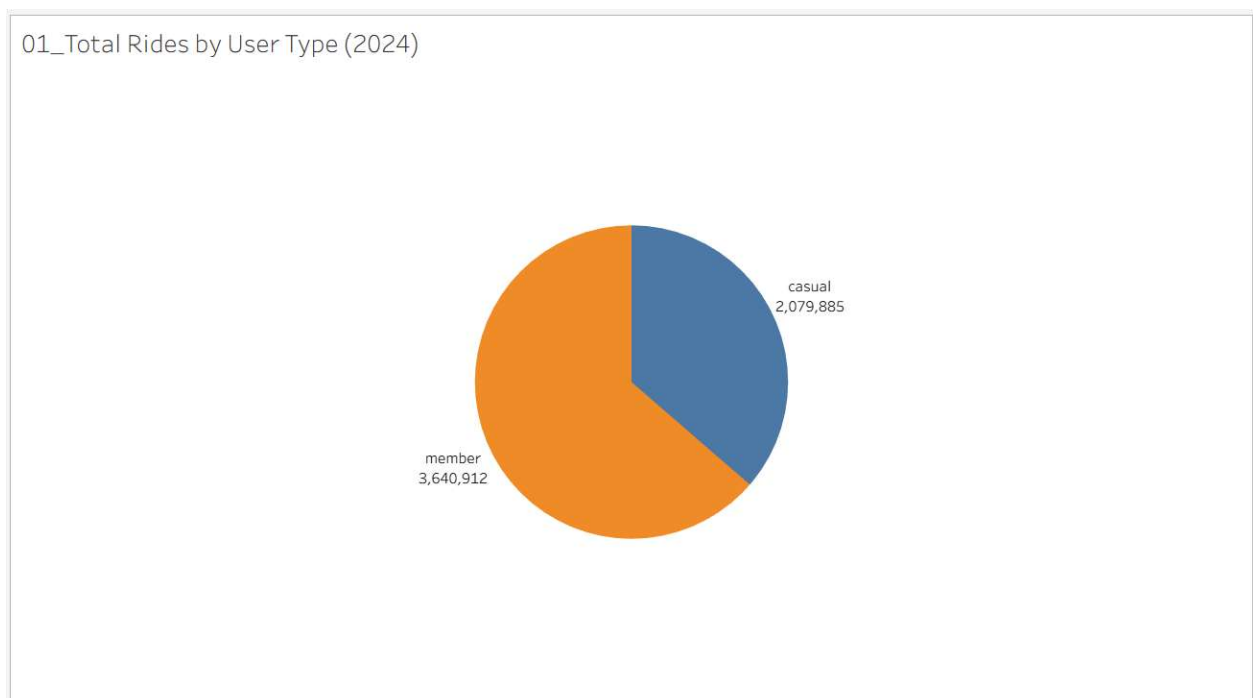
How do annual members and casual riders use Cyclistic bikes differently?

The cleaned data is imported into Tableau for analysis and the figures plotted are displayed in the following.

Total Rides in 2024:

The figure below shows the total number of rides carried out by Cyclistic members and casual riders in 2024.

- Total Rides by User Type (2024) (Pie Chart)



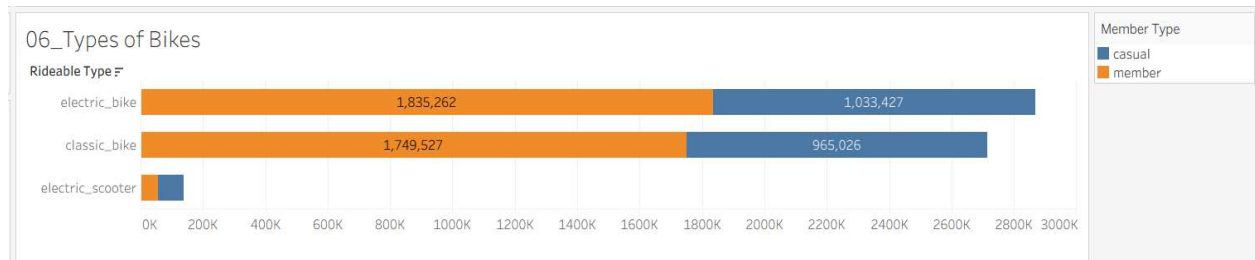
- Cyclistic members recorded greater bicycle activity than casual riders. The total rides for Cyclistic members are 3,640,912 while 2,079,885 trips for casual riders.

- Cyclistic members accounted for about 63.64% of total rides whereas casual riders made up 36.36% of total rides in 2024.

Types of Bikes:

The types of bicycles used for the trips are displayed as follow:

- Types of Bikes:

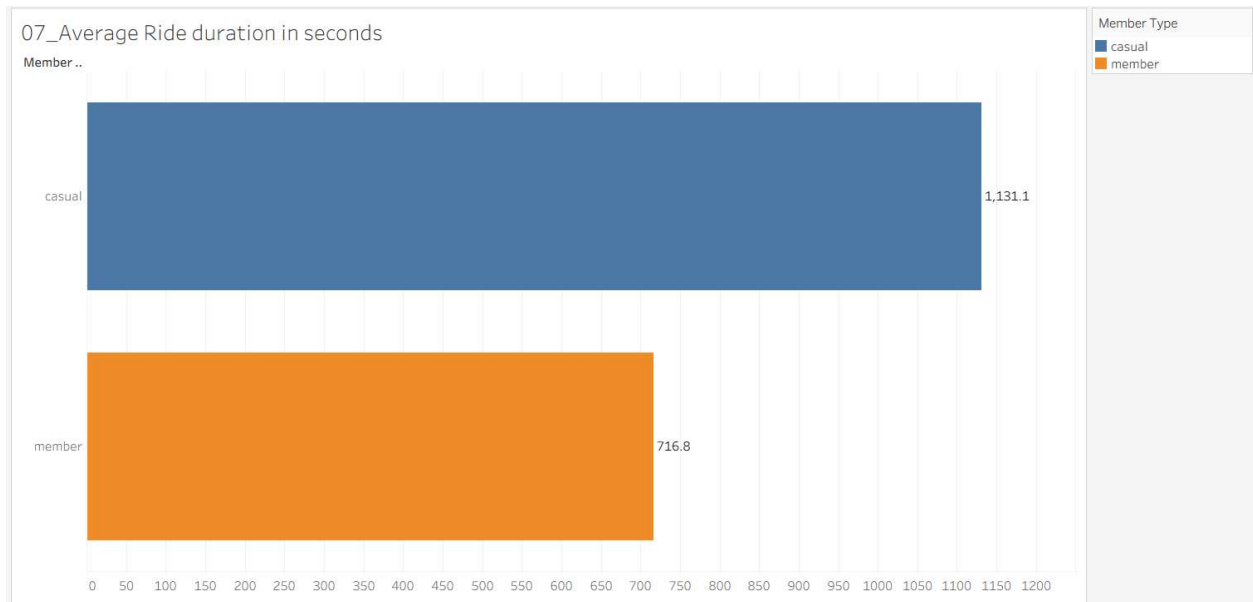


- There are three types of bicycles: *classic bikes*, *electric bikes*, and *electric scooters*.
- Cyclistic members and casual riders show a higher preference for electric bicycles over classic bicycles.
- Both casual riders & members have also used electric scooters.
- Members took 56,123 trips with electric scooters and casual users took 81,432 trips with electric scooters. (Note: Labels would have overlapped otherwise, which is why I mentioned the numerical details for electric scooters separately).

Average Ride Duration:

The average ride length is plotted against the type of users (member vs. casual):

- Average Ride duration in seconds (Member vs Casual user):

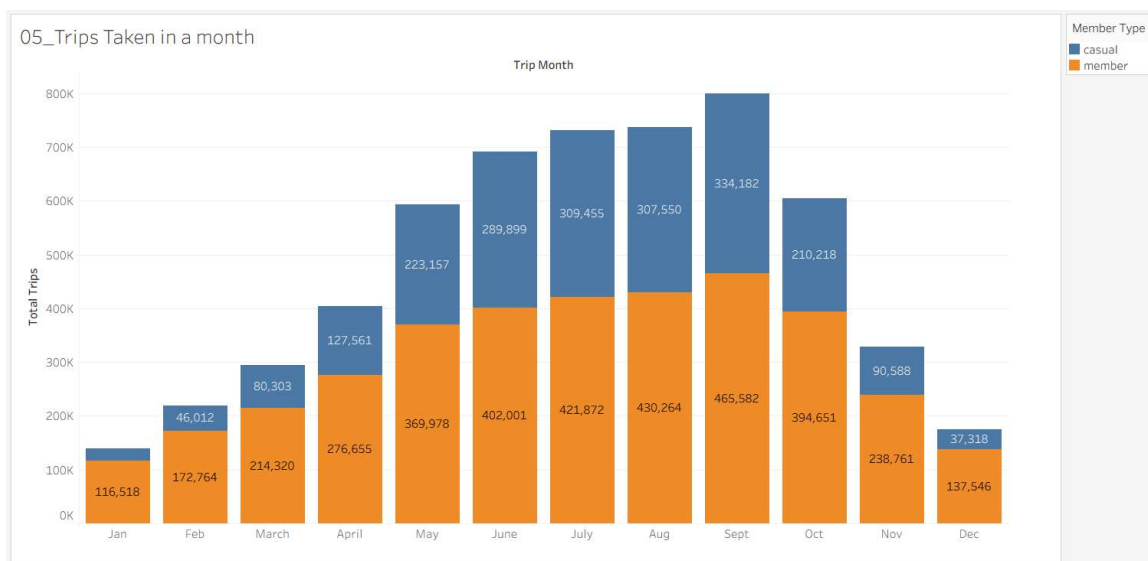


- Cyclic members can ride on the bicycles for about 716.8 seconds (11.94 mins) on average whereas casual riders have an average ride length of 1131.1 seconds (18.85 mins). Hence, the ride duration of Cyclic members are approximately 1.57 times smaller than casual riders.

Trips Taken in a Month:

The preference of cycling activity can be determined by drawing the graph of trips taken against the month from January to December 2024.

- Trips Taken in a month:

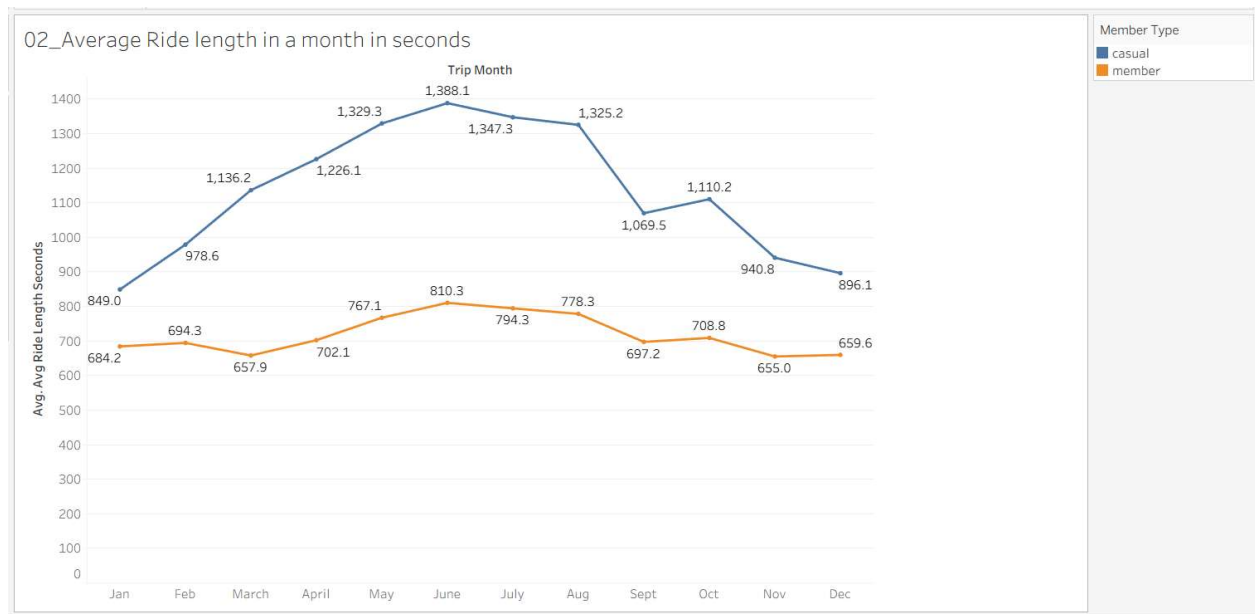


- Cyclistic members and casual riders have the lowest activity, 116,518 rides and 23,642 rides respectively in January 2024.
- Casual riders have the highest activity (334,182 rides) in September 2024 and Cyclistic members also have the greatest activity (465,582 rides) in July 2022.

Average Ride Length in a Month:

The mean trip duration is depicted in the line graph below.

- Average Ride length in a month (Line chart):

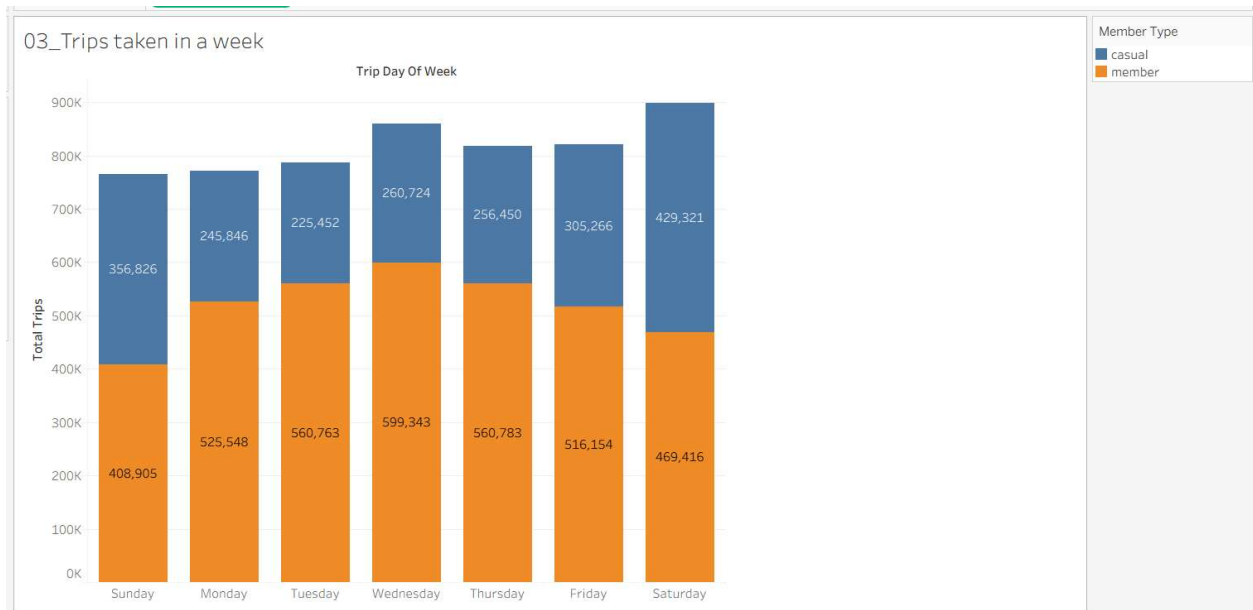


- The monthly average ride duration for Cyclistic members is the highest in June at 810.3 seconds, which is equal to 13.50 minutes.
- For casual riders, the highest mean trip duration is also in June at 1,388.1 seconds, which is equal to roughly 23.13 minutes.

Trips Taken in a Week:

The bar chart below is used to study the daily user activity over a week:

- Trips taken in a week (Stacked Column Chart)

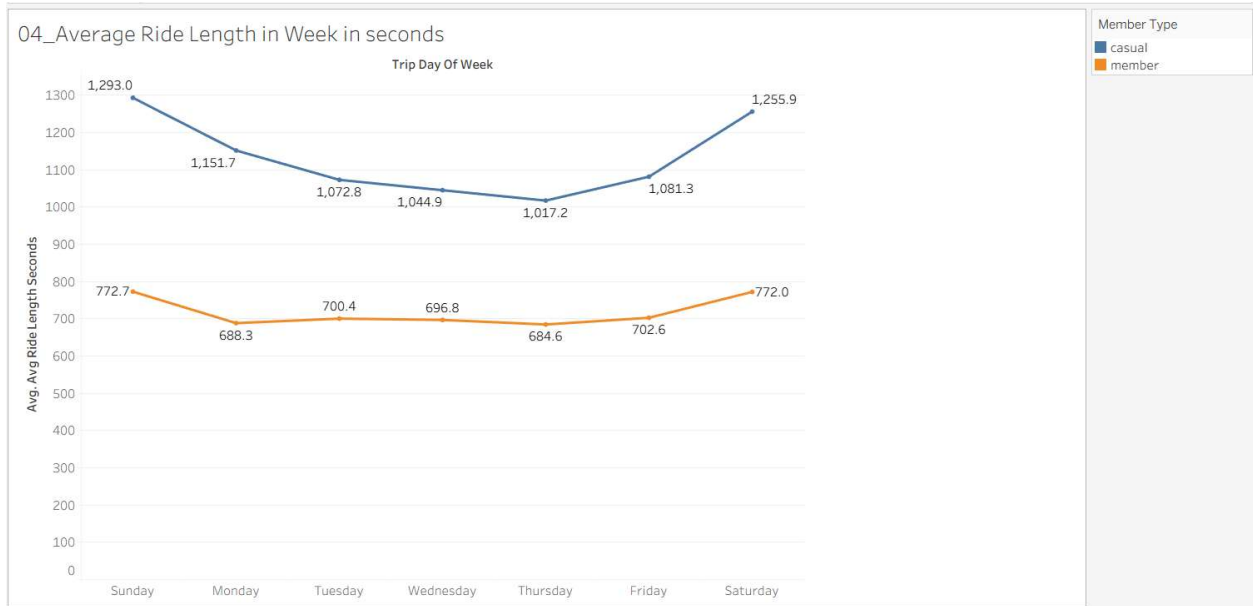


- Generally, bike rides are most frequented on Saturdays.
- Cyclistic members have the highest activity (599,343 rides) on Wednesdays while the lowest activity (408,905 rides) on Sundays.
- Casual riders have the greatest activity (429,321 rides) on Saturdays while the least activity (225,452 rides) on Tuesdays.

Average Ride Length in a Week:

The mean ride duration across the week is displayed as follows:

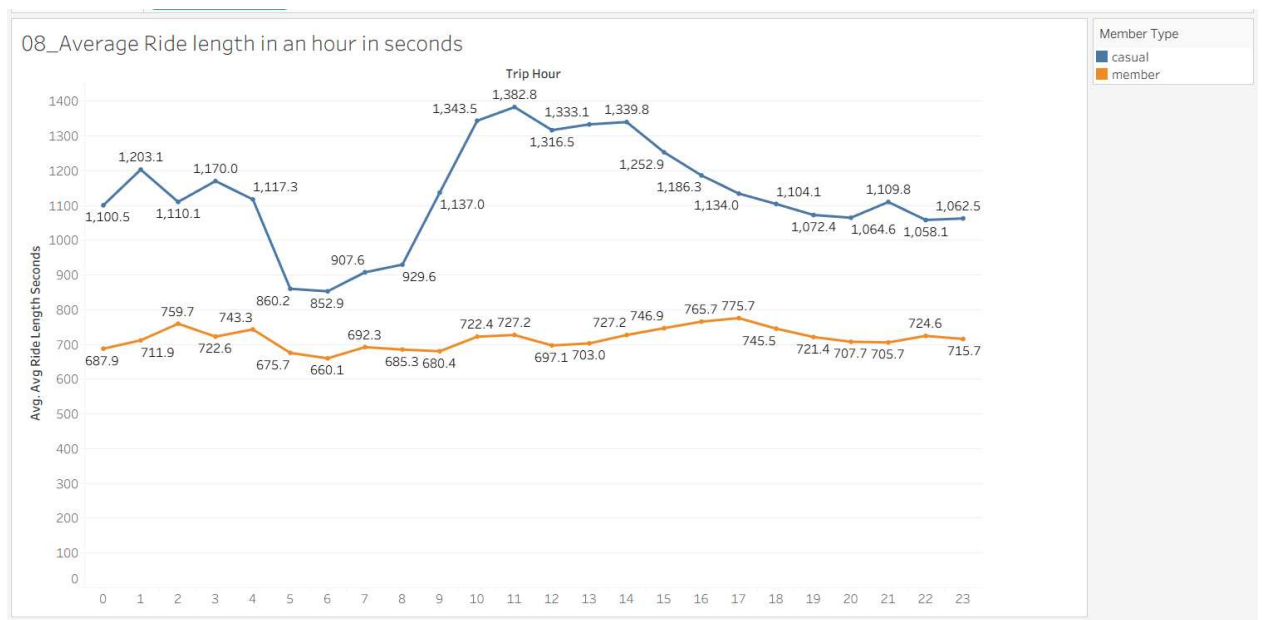
- Average Ride length in a week (Line chart)



- Cyclistic members cycled the longest on Saturday with an average ride length of 772 seconds (12.86 minutes),
- On the other hand, casual riders cycled the longest on Sunday with a mean trip duration of 1293 seconds (21.55 minutes).

Average Ride Length in an Hour:

- Average Ride Length in an Hour:



- As we can clearly see, casual members took the longest rides at 11 AM with 1382.8 seconds (23.04 minutes). Their shortest rides took place at 6 AM with 852.9 seconds (14.21 minutes).
- Members took their longest rides at 17 (17:00 or 5 PM) with 775.7 seconds (12.92 minutes) but their shortest rides were also at 6 AM with 660.1 seconds (11.001 minutes).

5. Share

In this phase, I have consolidated the insights into comprehensive Tableau dashboards to visualize the behavioral differences between members and casual riders.

Key Findings from the Dashboard

- **Market Share & Volume:** Cyclistic members recorded significantly greater activity, accounting for **63.64%** of total rides (3,640,912), while casual riders made up **36.36%** (2,079,885) in 2024.
- **Intensity of Use:** On average, casual riders have a much higher ride length of **1131.1 seconds** (~18.85 mins) compared to **716.8 seconds** (~11.94 mins) for members. This indicates that member rides are approximately **1.57 times shorter**, suggesting a "point-to-point" utility usage.
- **Seasonal & Monthly Trends:** Both groups have the lowest activity in **January**. Activity surges in the summer, with casual riders peaking in **September** (334,182 rides) and members peaking in **July** (465,582 rides).
- **Weekly Behavior:** Members show consistent activity during the week, peaking on **Wednesdays** (599,343 rides), while casual riders are most active on **Saturdays** (429,321 rides).
- **Hourly Patterns:** Casual riders take their longest rides at **11 AM** (1382.8 seconds), whereas members reach their peak duration at **5 PM** (775.7 seconds).
- **Product Preference:** Both groups show a higher preference for **electric bicycles** over classic bicycles. Additionally, casual users utilized **electric scooters** for **81,432 trips**, outperforming the **56,123 trips** taken by members.

6. Act

Based on these findings, I recommend the following marketing strategies to convert high-potential casual riders into annual members.

Recommendations

1. **The "Summer Explorer" Tier (Membership Personalization):** Introduce shorter-term membership options (monthly or daily) specifically for the high-intensity summer peak. This caters to casual riders who use the service for long trips but may not yet be ready for a full annual commitment.

2. **Electric Fleet Incentives:** Since casual riders show a strong preference for electric bikes and scooters, offer **member-exclusive access** or discounted rates for the electric fleet to entice them to join.
3. **"Weekend Warrior" Loyalty Program:** Implement a loyalty points system where frequent weekend riders (the peak time for casuals) earn rewards or discounts toward an annual membership.
4. **Targeted Digital & Social Campaigns:** Launch social media advertisements during peak casual hours (**weekends at 11 AM**) and during the high-activity summer months (May–September). Use user-generated content to showcase the leisure benefits of membership.
5. **Member-Exclusive Recreational Events:** Organize group rides or "Urban Exploration" challenges during the summer peak. Making these events member-exclusive adds social and experiential value to the annual subscription for leisure-focused casual riders.

Conclusion

This analysis confirms that while members provide the core volume for Cyclistic, **casual riders represent the highest-intensity users** in terms of trip duration and weekend engagement. While members use the service as a utility for commuting, casual riders use it as a premium leisure experience. By shifting from broad awareness to targeted conversion strategies, with specific focus on **seasonal peaks, electric vehicle preferences, and leisure-time engagement**, Cyclistic can successfully transition high-potential casual riders into loyal, profitable annual members, ensuring the long-term growth and success of the program.